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RESEARCH ARTICLE

LegalBot-EC: An LLM-Based Chatbot for Legal Assistance in Ecuadorian Law

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ABSTRACT This work presents LegalBot-EC, a domain-specific legal chatbot designed to provide accurate responses grounded in Ecuadorian law. The system combines ChromaDB for contextual retrieval and LLaMA 3.1 as its generative model, ensuring legally relevant and contextually grounded answers. Its knowledge base consists of two key legal documents, the COIP and the 2008 Constitution of Ecuador, which serve as the foundation for response generation. The chatbot was evaluated through two key assessments: user satisfaction and accuracy measurement. A satisfaction study conducted with final-year university students specializing in law resulted in an average score of 88.72 out of 100, indicating strong approval in terms of correctness, relevance, and clarity. Additionally, an accuracy assessment based on 25 legal queries demonstrated robust performance, with 23 correct responses, 2 partially correct responses, and none classified as incorrect, yielding a weighted accuracy of 96%. Furthermore, qualitative tests confirm the chatbot's ability to generate well-structured, legally sound responses, even when queried in English, demonstrating multilingual capabilities.

INDEX TERMS Large language models, chatbot, natural language processing, artificial intelligence, legal informatics, legal AI, BERT, LLaMA, transformer.

I. INTRODUCTION

Modern legal systems are built upon a vast collection of statutes, codes, and regulations [1], [2]. However, the growing volume of these texts, combined with their highly technical and specialized language, poses a significant challenge to their comprehension [3]. This issue affects both citizens seeking guidance and legal professionals who require quick and accurate responses in increasingly complex scenarios [4]. The challenge becomes even greater when identifying specific provisions or interpreting intricate legal concepts [5]. In addition to the complexity of legal language, other factors exacerbate this issue. The high costs of legal consultation and representation limit litigants' access to proper advice [5], [6], particularly in low-income sectors.

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These challenges are compounded by limited access to up-to-date legal information and the lack of tools that facilitate efficient interaction with legal texts. Together, these obstacles not only restrict access to justice but also perpetuate inequalities in the exercise of fundamental rights.

The digitalization of information has brought about significant advancements in various domains, facilitating access to knowledge and services. In many cases, digital tools and systems have been developed to improve the accessibility and usability of complex datasets and documents [7]. However, the adoption of such systems remains limited in certain sectors, particularly in the legal domain, where the reliance on traditional practices often persists [8]. This limitation is even more pronounced in developing countries [9], where technological infrastructure may be unevenly distributed, and digital solutions often fail to address the specific needs of local contexts [4], [8]. These gaps underscore the necessity of

developing tailored systems that democratize access to legal information and enhance its usability. Such advancements are particularly critical for bridging the accessibility divide, ensuring that legal knowledge is not only available but also comprehensible and actionable for diverse audiences. By addressing these challenges, digital tools can play a pivotal role in promoting equal access to justice and fostering more inclusive legal systems [10]. Artificial Intelligence (AI) has emerged as a game-changing technology capable of addressing complex challenges in a variety of fields. It uses advanced computational models to process, analyze and interpret large amounts of data with high accuracy and efficiency [11]. Among the techniques within AI, Natural Language Processing (NLP) stands out as a fundamental tool for enabling machines to understand, interpret and generate human language [12]. NLP has seen significant advances in recent years, particularly in its ability to process complex linguistic structures and extract meaningful information from unstructured text [13], [14]. Because of this, NLP techniques have proven to be highly effective in tasks such as text analysis, language translation and contextual understanding [15]. A recent milestone within NLP is the development of Large Language Models (LLMs). These models are trained on massive datasets, [15], [16] enabling them to capture nuanced linguistic patterns and contextual relationships. As a result, LLMs can generate coherent, contextually relevant responses with human-like fluency and adapt to diverse tasks [17], [18], ranging from content creation to conversational interfaces [19]. Their versatility and scalability make them a cornerstone of modern NLP applications, offering unprecedented opportunities for innovation in human-computer interaction.

Building on the capabilities of LLMs, chatbots have become one of the most prominent applications of NLP. These interactive programs are designed to mimic human behavior or conversation [20], enabling users to engage with systems through natural language. By leveraging the contextual understanding and text generation capabilities of LLMs [19], [21], chatbots can now handle increasingly complex interactions with remarkable accuracy and fluency [22]. Among their many applications, chatbots stand out as a particularly promising tool in sectors such as healthcare [23], customer service [24], and, notably, the legal domain [25]. By bridging the gap between complex legal language and user-friendly interaction, chatbots address the lack of tools that facilitate efficient engagement with legal texts [26]. Their ability to provide direct access to relevant information in a conversational format simplifies the navigation of intricate legal documents and datasets [15], [26], making them a valuable resource for improving accessibility and understanding, and driving the transformation of traditional legal practices.

Building on the ideas presented above, this work introduces a chatbot designed to assist users in navigating and understanding legal texts. Specifically, we focus on two fundamental components of Ecuador's legal framework:

the Comprehensive Organic Criminal Code (COIP) and the Constitution. The COIP consolidates criminal laws and procedures, while the Constitution serves as the supreme legal document, outlining the fundamental principles and rights of the nation. Both texts, due to their technical language and extensive structure, often present challenges for legal professionals and the general public alike [4]. The chatbot leverages Large Language Model Meta AI (LLaMA), a state-of-the-art LLM, as its main architecture. By tailoring this model to the specific contexts of the COIP and the Constitution, we aim to provide an interactive and accessible tool that bridges the gap between users and complex legal information, offering a practical solution for improving engagement with Ecuador's legal system. To the best of our knowledge, this is the first chatbot specifically developed for legal assistance in Ecuador, representing a novel step toward the digitalization and accessibility of the country's legal resources.

II. RELATED WORK

The development of chatbots for legal assistance has been an area of growing interest in recent years. Several studies have explored the design and implementation of such systems. These works range from general-purpose legal chatbots to highly specialized systems tailored to specific legal issues. This section reviews notable works in this area, examining their design, scope, and contributions to the advancement of legal chatbot technologies.

To begin with, in [27], a chatbot is developed to provide answers to user questions about Islamic law, specifically focusing on religious practices such as prayer (shalat) and zakat. The chatbot uses a dataset named "data.rive," which consists of question-answer pairs sourced from Islamic websites. The system processes user inputs in natural language and retrieves relevant answers based on Islamic texts, such as verses from the Qur'an or Hadith. The chatbot employs a two-step approach for processing queries. First, stemming is performed to normalize words by removing affixes and converting them into root forms using a dictionary-based approach. Second, the system applies a fuzzy string-matching algorithm to compare the processed input with existing entries in the dataset. If a match is found, the chatbot retrieves the corresponding answer from the question-answer pair. If no direct match is available, further checks on word patterns, string length, and positional similarity are conducted to identify the closest match.

Similarly, the authors in [28] developed a chatbot to provide legal assistance for understanding Indonesia's Electronic Information and Transaction Law (ITE Law No. 19 of 2016). The system was designed to handle user queries about legal documents using a NLP pipeline, including components such as a parser, lexicon, understander, knowledge base, and generator. The chatbot processes user inputs through multiple stages, starting with parsing the query and correcting non-standard words using the Levenshtein distance method to ensure compatibility with the system. Once the input is

normalized, the chatbot employs TF-IDF cosine similarity to compare the processed query with existing questions and legal documents in the database. The most relevant match is then retrieved and used to generate a response tailored to the user's input. In the context of Thai law, [29] presents a chatbot to assist survivors of sexual violence by recommending relevant Supreme Court decisions. The system focuses on Sections 276-279 of the Thai Criminal Code and utilizes a dataset of 182 related court cases. Through an NLP pipeline, the chatbot processes user input, extracting key information about the incident. The chatbot evaluates three metrics: similarity scores derived from word embeddings, the percentage of common keywords, and the percentage of common keyword synonyms. Based on these calculations, the system recommends the three court cases most closely aligned with the user's situation, providing survivors with a starting point for understanding their legal standing.

In [30], a prototype chatbot is introduced to support victims of Non-consensual Intimate Images (NCII) violence in reporting their cases and accessing justice. The chatbot aims to improve the legal quality of abuse narratives by guiding users through an interactive process that collects relevant information and offers tailored recommendations. These recommendations can include detailed procedural descriptions, customized advice, or pre-formatted text suggestions for submission during the reporting process. The system adopts a declarative flow-based approach, leveraging domain-specific properties and rules to dynamically guide the user. The chatbot relies on a decision tree architecture, where each interaction node gathers specific information from the user, influencing the subsequent path of the conversation. Domain experts define properties and rules that determine how decisions are reached, while an automatic algorithm generates the decision tree from the declarative specifications.

Deep learning-based solutions have gained prominence in recent years, as exemplified in [31], where a chatbot is developed to assist users in navigating legal concepts and procedures, with a specific focus on conflict resolution within the European Union. The chatbot provides guidance on disputes related to divorce, inheritance, or company division. The system aims to help users navigate legal terminology, understand procedures, and receive recommendations tailored to specific conflict scenarios. It employs Sentence-BERT (SBERT), a transformer-based model optimized for generating semantically meaningful sentence embeddings. Using a dataset of over 300 question-answer pairs crafted by legal professionals, the chatbot matches user queries with the most similar questions in its knowledge base. By leveraging semantic search techniques, the chatbot identifies and retrieves the closest question, providing the associated answer as guidance.

Following a similar trend, [32] presents a chatbot designed to address legal questions related to Palestinian cooperative law, with a specific focus on regulations outlined in Law No. 20 of 2017, Cooperatives Bylaws, and Housing Cooperatives Bylaws. The system combines both human-generated

and AI-generated datasets to build its knowledge base. The human-generated dataset was created with the assistance of legal advisors, while the AI-generated dataset utilized ChatGPT to produce complementary question-answer pairs. The chatbot employs LlamaIndex as an indexing and retrieval mechanism. Legal documents are indexed to generate vector representations. When a user inputs a legal question, the chatbot formulates a query by matching the input with the indexed vectors. The query is processed by GPT-4 to generate an accurate and context-specific legal response.

The analysis of the reviewed works highlights the growing interest in leveraging chatbots for legal assistance across diverse legal systems and contexts. These efforts demonstrate how NLP and, more recently, advanced techniques like deep learning, have been employed to bridge the gap between legal texts and user understanding. While the approaches vary in complexity, they share a common goal of improving accessibility to legal information and guidance.

However, the geographic and thematic scope of these efforts remains limited. Most of the systems focus on legal frameworks specific to countries in Asia, Europe, or the Middle East, with little to no exploration of applications tailored to Latin American contexts. This is consistent with the findings in [3], which stated that most existing research emphasizes international legal texts, particularly Asian and European cases. Furthermore, when focusing on Ecuador, the absence of solutions or chatbots designed for its legislation becomes evident, highlighting a significant gap in this field.

III. OVERVIEW OF ECUADOR'S LEGAL FRAMEWORK

A. COMPREHENSIVE ORGANIC CRIMINAL CODE

The Comprehensive Organic Criminal Code (Código Orgánico Integral Penal, COIP) is a cornerstone of Ecuador's legal framework, enacted in 2014 to modernize the country's criminal justice system. It succeeded the previous Criminal Code, which had been in force since 2000 and faced widespread criticism for being outdated and inadequately addressing the evolving nature of crime and societal values. The COIP was designed to align Ecuador's legal provisions with international norms and standards, reflecting a more contemporary understanding of justice and criminal behavior.

The COIP provides a unified legal foundation for addressing criminal offenses in Ecuador, outlining the rights and responsibilities of individuals within the justice system and establishing guidelines for judicial procedures. It encompasses over 700 articles, from general principles of criminal law to specific regulations on offenses, sentencing, and procedural rules. By integrating these elements into a unified code, the COIP aims to ensure consistency and fairness in the application of justice, while also providing clear standards for legal practitioners and citizens alike.

Since its enactment, the COIP has undergone two major updates, in 2021 and 2023, to address emerging challenges and adapt to new legal and societal demands. However, its structure remains consistent, divided into a preliminary book

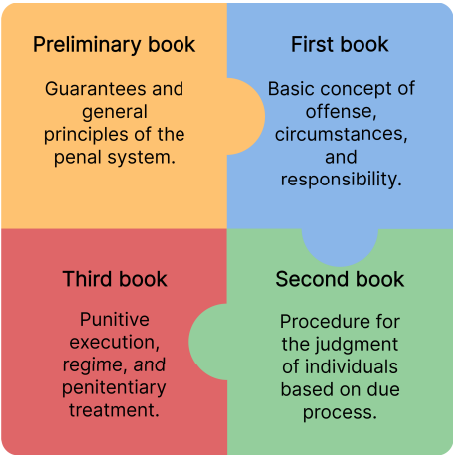


FIGURE 1. Structure of the Ecuadorian comprehensive organic criminal code.

and three subsequent books, as illustrated in Figure 1. These components collectively provide a comprehensive framework for Ecuador’s criminal justice system, each serving a specific purpose:

- The preliminary book outlines the guiding principles and foundational norms that serve as the theoretical framework for the code. It sets the philosophical and legal foundation, ensuring coherence with the overarching objectives of the Ecuadorian justice system.
- The first book focuses on defining and categorizing criminal offenses. It provides an exhaustive classification of crimes, ranging from minor infractions to severe felonies, alongside the legal definitions and parameters for each.
- The second book addresses procedural aspects, covering every stage of a legal case, from initiation and investigation to adjudication and appeals. This section offers detailed guidance for legal actors, including judges, prosecutors, and defense attorneys, ensuring due process and fairness.
- The third book elaborates on the execution of penalties. It details the practical considerations for administering sentences, whether they involve incarceration, alternative sanctions, or non-custodial measures, emphasizing the principles of rehabilitation and proportionality.

The COIP encompasses a wide array of criminal offenses, including crimes against individuals, property, public order, and specialized areas such as cybercrime and environmental law. Additionally, it integrates principles of proportionality and rehabilitation within its sentencing framework, reflecting its dual focus on justice and reintegration.

B. 2008 CONSTITUTION OF ECUADOR

The Constitution of Ecuador, enacted in 2008, is the highest legal authority in the country, providing the foundation for all other laws and regulations. It establishes the principles that govern the state and defines the rights and responsibilities of its citizens and institutions. As the supreme legal framework,

Preamble Affirms sovereignty, recognizes cultural diversity, and establishes "sumak kawsay" (good living) as a guiding principle.	Title I: Constitutive elements of the state Defines Ecuador as a democratic, secular, and sovereign republic, outlining state responsibilities and citizenship criteria.	Title II: Rights Lists individual rights, including health, education, and environmental protections, while addressing vulnerable groups and nature.	Title III: Constitutional guarantees Ensures enforcement of rights through mechanisms protecting various societal groups.
Title IV: Participation and organization of power Outlines citizen participation and the five state functions: legislative, executive, judicial, electoral, and transparency.		Title V: Territorial organization of the state Divides the country into regions, provinces, cantons, and rural parishes, governed by decentralized authorities.	Title VI: Development regime Establishes a framework for sustainable political, economic, social, and cultural growth.
Title VII: Good living regime Focuses on equity, inclusion, health, education, and biodiversity.	Title VIII: International relations Advocates for universal citizenship, free mobility, and equitable international cooperation.	Title IX: Supremacy of the constitution Declares the Constitution as the highest legal authority, overriding any conflicting laws.	

FIGURE 2. Structure of the Ecuadorian constitution of 2008.

the Constitution takes precedence over all other legal norms, including international treaties, organic and ordinary laws, and regulations issued by decentralized or public institutions, except in cases where international human rights standards provide greater protection.

The 2008 Constitution of Ecuador consists of a preamble and nine titles, as shown in Fig. 2, covering a total of 444 articles. A notable distinction from previous constitutions is its innovative separation of powers, which extends beyond the traditional branches, executive, legislative, and judicial, to include two additional functions: the electoral function, responsible for overseeing democratic processes, and the transparency and social control function, tasked with promoting accountability and preventing corruption.

One of the defining characteristics of the 2008 Constitution is its progressive approach to rights and social issues. It was the first Constitution in the world to recognize legally enforceable Rights of Nature, granting ecosystems the right to exist, flourish, and be restored. Additionally, it acknowledges the right to food as a fundamental human right and frames drug consumption not as a criminal offense but as a public health concern. The Constitution also made significant advances in recognizing the rights of LGBTQ+ individuals, including the recognition of same-sex civil unions and the explicit inclusion of sexual orientation and gender identity as protected categories under fundamental rights.

Since its enactment, the Constitution has undergone three modifications through two reforms and one amendment in 2011, 2014, and 2018. These changes addressed key issues such as the indefinite reelection of public officials, adjustments to the administration of justice, and revisions to deadlines in criminal proceedings. These updates underscore the dynamic nature of the Constitution, reflecting ongoing efforts to adapt its provisions to the evolving political and social landscape of Ecuador.

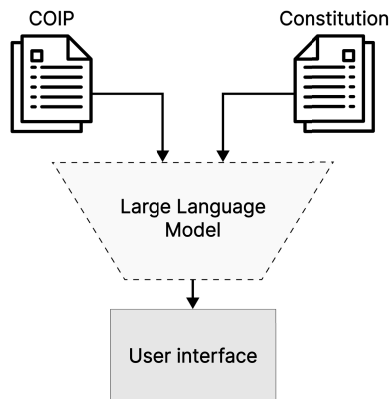


FIGURE 3. High-level diagram of the chatbot developed in this work.

IV. CHATBOT DEVELOPMENT

At a high level, the chatbot system integrates multiple components, as illustrated in Fig. 3. The system begins with the ingestion of legal texts, specifically the COIP and the 2008 Constitution of Ecuador, which form the core of its knowledge base. Users interact with the chatbot through natural language queries, and the system retrieves the most pertinent legal text and generates a response. The response is presented in a user-friendly format, ensuring accessibility even for individuals without legal expertise. Initially, the chatbot is tailored specifically for Ecuadorian users. As the target audience primarily speaks Spanish, the chatbot was developed in this language to ensure accessibility and usability for its intended users. In the following subsections, we delve into the details of the system's development.

A. DATA COLLECTION

The first step in developing the chatbot involves collecting and processing legal documents, specifically Ecuador's COIP and the 2008 Constitution. These documents are primarily stored in PDF format. To extract text from these files, the PyMuPDF¹ library was used. PyMuPDF is a Python-based tool designed for efficient data extraction and manipulation of PDF documents.

The extraction process retrieves content from all pages of each document, focusing on capturing all relevant content from the documents, such as articles, sections, and clauses, while discarding formatting elements or metadata that do not contribute to the chatbot's understanding. The cleaning steps included the following:

- **Elimination of irrelevant elements:** Metadata, page numbers, headers, footers, and any duplicated sections were removed to reduce noise.
- **Normalization of special characters:** Issues such as line breaks within words, hyphenation across lines, and inconsistent spacing were addressed to ensure coherence in the textual data.
- **Formatting adjustments:** The text was reformatted to preserve the original semantic structure, including

maintaining article titles, sections, and subsections in a logical order.

- **Exclusion of non-textual content:** Decorative elements, improperly parsed tables, and graphics that could not be interpreted as text were discarded.

These steps ensured that the extracted text was semantically coherent, preserving critical legal content while removing extraneous information that could interfere with downstream tasks. The cleaned and processed text serves as the foundation for the embedding creation stage, where its semantic representation is generated. By maintaining a structured and high-quality dataset, this stage provides a robust input for building the chatbot's knowledge base and ensuring the relevance of its responses.

B. EMBEDDING CREATION

To enable the system to efficiently process and comprehend legal information, the extracted text is transformed into embeddings. An embedding is a way of representing objects such as text, images, or audio as points in a continuous vector space [33], [34]. In the context of NLP, embeddings are dense numerical vectors that capture the semantic meaning of text [33], [35]. They encode linguistic and contextual relationships within the data [36], enabling nuanced comparisons between text fragments and capturing intrinsic similarities. This allows artificial intelligence models to learn complex patterns and relationships from the data [37]. For textual data, embeddings are typically generated by passing sentences, paragraphs, or even entire documents through a neural network model trained on large corpora [38]. The model processes the input text and outputs a low-dimensional vector that encapsulates its semantic characteristics [39]. These embeddings form the foundation of the chatbot's knowledge base, facilitating the matching of user queries with the most relevant legal content.

In this work, BETO [40] is employed as the embedding model to generate semantic representations of the legal text. BETO is a Spanish-adapted version of the Bidirectional Encoder Representations from Transformers (BERT) [41], shown in Fig. 4, a highly influential architecture in NLP. The key innovation of BERT lies in its bidirectional training of Transformer models [42]. Unlike previous approaches that analyzed text sequences unidirectionally (either left-to-right or a combined left-to-right and right-to-left approach) [43], BERT examines a word in the context of the words surrounding it. This bidirectional understanding allows the model to capture richer contextual relationships [44], significantly improving its ability to comprehend and process language. As a result, BERT has achieved remarkable performance across a wide range of NLP tasks [45], and BETO extends this capability to the Spanish language.

BETO was trained on a large Spanish-language corpus comprising approximately 3 billion words from diverse sources, including Wikipedia, United Nations and government journals, TED Talks, subtitles, and news articles [40]. Its

¹<https://pymupdf.readthedocs.io/en/latest/>

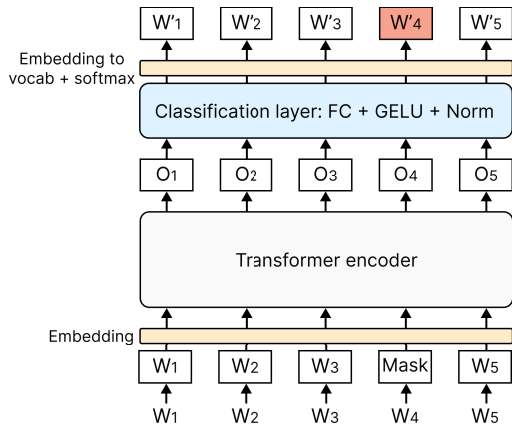


FIGURE 4. BERT architecture.

architecture mirrors that of BERT-Base, with 12 self-attention layers, 12 attention heads per layer, and a hidden size of 768, totaling 110 million parameters [46], [47]. The model was further fine-tuned using a whole-word masking technique [48], enhancing its ability to predict and understand complete word representations in context.

The embeddings are generated by passing the cleaned legal text through BETO. The model processes the input and outputs a dense semantic representation from its last hidden state layer. This representation captures the contextual and linguistic meaning of the text, forming the basis for the chatbot's ability to retrieve and match relevant legal content efficiently. To facilitate efficient retrieval and indexing, the generated embeddings, along with the corresponding extracted text from the legal documents, are stored in ChromaDB.² ChromaDB is a specialized vector database designed to manage high-dimensional embeddings, enabling fast and accurate similarity searches, as shown in Fig. 5. Each embedding is directly linked to its associated legal text from the PDFs, creating a searchable collection where user queries can be matched with the most relevant content based on semantic proximity. For this work, BETO was sourced from the Hugging Face library.³

C. ANSWER REFINEMENT

Large Language Model Meta AI (LLaMA) 3.1 is an LLM released by Meta, representing one of the most advanced open-source NLP models. It builds on the foundations laid by its predecessors, LLaMA 1, 2, and 3, incorporating an increased number of parameters and a larger, more diverse training corpus, designed to enhance its capacity for natural language understanding and generation. The LLaMA series is built upon the Transformer architecture [49], which serves as its foundational framework. Introduced in 2017, the Transformer revolutionized NLP with its ability to process entire sequences of text in parallel, rather than sequentially. This architectural shift significantly improved the efficiency

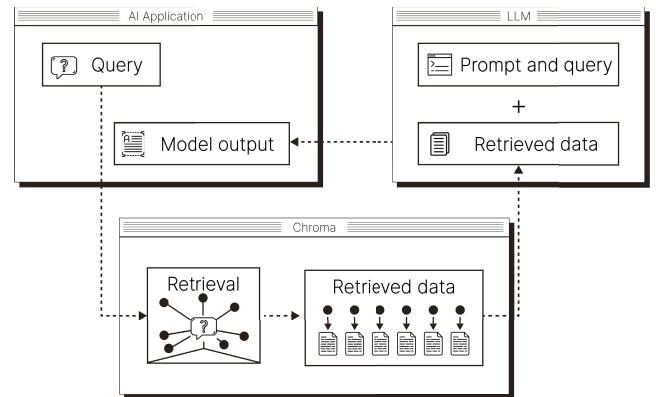


FIGURE 5. Operation of an AI application using chroma database.

and scalability of deep learning models for NLP tasks, allowing them to handle larger datasets and more complex language structures.

At the heart of the Transformer, shown in Fig. 6, lies the self-attention mechanism, a key innovation that allows the model to evaluate the importance of each word in a sequence relative to all others. By calculating attention scores, the Transformer can capture contextual relationships effectively, which is crucial for understanding dependencies across long stretches of text. This mechanism is implemented through multi-head attention modules, which compute multiple attention distributions in parallel, enabling the model to focus on different aspects of the input simultaneously. Complementing this is the feedforward neural network, which processes the output of the attention layers to introduce non-linearity. Moreover, to encode the positional order of tokens, the Transformer uses positional encodings, which represent each token's position through sinusoidal functions, ensuring the model retains information about sequence order despite parallel processing [50]. Finally, layer normalization is applied throughout the architecture to stabilize training and improve convergence. Together, these components enable Transformers to excel in capturing complex linguistic patterns and dependencies, positioning them as a cornerstone of modern NLP models.

LLaMA 3 builds on the architecture of its predecessors, incorporating refinements to improve efficiency and performance. LLaMA employs only the decoder component of the Transformer architecture, as shown in Fig. 7, aligning with its design for next-token prediction tasks. One of the key enhancements in LLaMA is the use of Root Mean Square Normalization (RMS Norm). RMS Norm is a simplified and efficient alternative to standard Layer Normalization [51]. Unlike Layer Normalization, which normalizes features based on their mean and variance, RMS Norm relies solely on the root mean square of the activations. This approach eliminates the need to compute the mean and variance [51], [52], thereby reducing computational complexity and enhancing efficiency.

LLaMA 3 also replaces traditional positional encodings with Rotary Positional Encodings (RoPE). Unlike fixed

²<https://www.trychroma.com/>

³<https://huggingface.co/dccuchile/bert-base-spanish-wwm-cased>

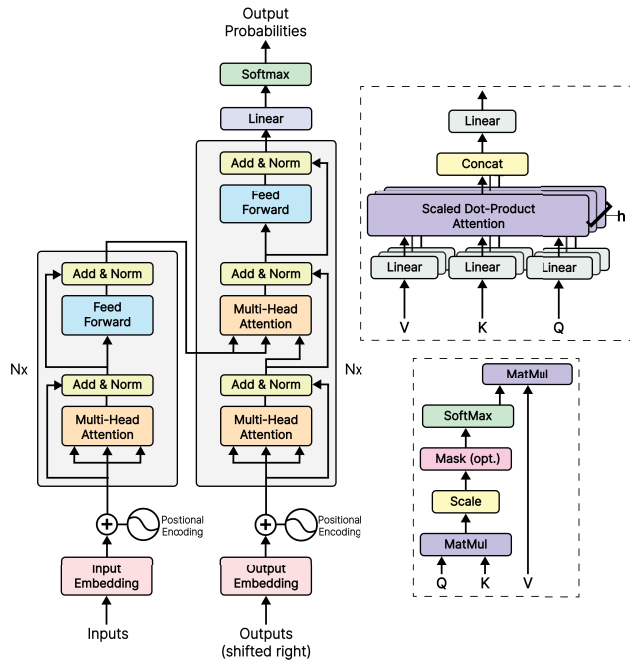


FIGURE 6. Transformer architecture and its components. Taken from [50].

absolute encodings, RoPE represents the relative distance between tokens by applying rotational transformations directly to the Query and Key vectors in the attention mechanism [53], [54]. This approach improves the model's ability to encode contextual dependencies while maintaining computational efficiency. Furthermore, RoPE naturally incorporates long-term decay [54], reducing the influence of distant tokens, helping the model focus on relevant local relationships without losing the ability to process long-range dependencies.

Moreover, LLaMA 3 incorporates Grouped Multi-Query Attention (GMQA) to enhance memory efficiency and computational performance. Traditional self-attention requires recalculating the attention scores for all tokens in the input sequence at each step, leading to computational redundancy [55], [56]. The Key-Value (KV) cache addresses this by storing the Key and Value matrices computed in previous steps. During inference, only the Query matrix for the new token is computed [57], while the cached Key and Value matrices are reused. This approach significantly reduces the computational burden by avoiding redundant calculations for already processed tokens [58]. Furthermore, the KV cache maintains the causal nature of the model by ensuring that each token only attends to its predecessors [56], aligning with the next-token prediction task for which LLaMA models are designed.

Finally, LLaMA 3 replaces the standard activation function in its feedforward layers with SwiGLU, a variant of the Gated Linear Unit (GLU). SwiGLU combines the Swish activation function with GLU [51], using Swish in place of the traditional sigmoid function to introduce non-linearity. This

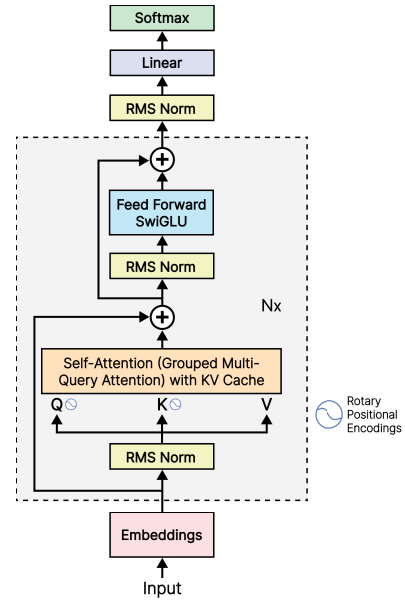


FIGURE 7. LLaMA 3 architecture.

design enhances the expressiveness of the model and reduces training times by accelerating convergence [59]. LLaMA 3.1 builds on LLaMA 3 with targeted improvements designed to enhance its usability and performance. Released shortly after LLaMA 3, this version features an expanded context window and increased maximum output tokens, enabling it to process longer and more complex tasks effectively. These updates make LLaMA 3.1 more adaptable to a variety of applications. LLaMA 3.1 is offered in three configurations: 8B, 70B, and 405B parameters. For this study, we selected the 8B variant as it balances performance and computational efficiency, aligning well with the project needs.

In our chatbot, LLaMA 3.1 serves as the core generative model, integrated with a retrieval-augmented generation (RAG) pipeline to provide context-aware and enriched responses. This approach ensures that the chatbot leverages both the semantic knowledge embedded in the legal documents and LLaMA's generative capabilities. The workflow operates as follows:

- **Contextual retrieval:** When a user submits a query, the system first performs a semantic search within ChromaDB, which stores the vector representations of legal texts obtained during the embedding creation phase. These embeddings represent fragments of the documents, such as articles, sections, and clauses, and are retrieved based on their relevance to the user's query. This step ensures that only the most pertinent content is passed to the model.
- **Context injection:** The retrieved text fragments from ChromaDB are injected into the input prompt for LLaMA 3.1. This process acts as contextual grounding, where the retrieved fragments provide a focused legal context that guides the model's response generation. By incorporating this context into the input, the model

ensures that its responses are directly informed by the specific legal context retrieved from ChromaDB.

- **Response generation:** LLaMA 3.1 then synthesizes a response by combining the provided context with its own pre-trained knowledge. Rather than merely extracting or repeating text, the model reformulates and enriches the information, ensuring the response is accurate, clear, and tailored to the user's question. Additionally, when the retrieved fragments are insufficient or incomplete, LLaMA supplements the response using its generalized knowledge, maintaining relevance to the legal domain.

In summary, the chatbot integrates semantic retrieval from ChromaDB with the generative capabilities of LLaMA to provide context-aware responses. The retrieved legal text fragments guide the model's response generation, ensuring answers are grounded in the uploaded legal documents while supplementing with pre-trained knowledge when necessary. This combination enables the chatbot to handle user queries efficiently and accurately. Fig. 8 illustrates the complete workflow of the chatbot, showcasing the interaction between its key components.

D. APPLICATION IMPLEMENTATION

Once the flow and architecture of the chatbot were defined, the next step was to implement it as a functional application. This stage emphasized creating a straightforward yet effective design, balancing usability with technical capabilities. The application integrates a front-end interface where users can interact with the system and a back-end infrastructure that manages query processing, contextual retrieval, and response generation. The focus remained on maintaining simplicity to accommodate the target audience while leveraging advanced tools to deliver reliable performance.

For the backend of this project, Ollama⁴ was used to manage and run LLMs locally. This tool provides an isolated environment for deploying AI models directly on local machines, ensuring that all processing occurs securely and privately without relying on external servers. Ollama's environment includes essential components such as pre-trained weights, configuration files, and necessary dependencies, simplifying the deployment process. It enables seamless integration with LLMs models, which are loaded and executed via the Ollama API to support contextual response generation. Additionally, Ollama offers compatibility across major platforms like Linux, macOS, and Windows, providing adaptability to various development workflows.

For the frontend of this project, Open WebUI⁵ was used as the interface to facilitate user interaction with the chatbot. Open WebUI is a self-hosted, extensible, and user-friendly graphical user interface (GUI) designed to simplify interaction with LLMs, while operating entirely offline. Its primary role in this project is to provide an intuitive platform through which users can submit queries and receive

responses from the chatbot. A key advantage of Open WebUI is its flexibility. It supports local deployment, ensuring that all data and interactions are stored and processed on the user's device. This approach prioritizes privacy and data security, an essential aspect when handling sensitive legal queries. The interface is designed to accommodate workflows requiring context-specific interactions, such as uploading documents or referencing external URLs, enabling the chatbot to deliver more tailored and relevant responses.

The interface is organized into three main sections. The top panel provides controls for managing the chat session, selecting AI models, and accessing advanced settings. The left sidebar offers options for navigating chat history, starting new sessions, or managing user workspaces. At the core of the interface is the main chat window, which displays the user's queries and the chatbot's responses, as shown in Fig. 9. The integration of Open WebUI ensures a seamless connection between the intricate backend processes and the user, enabling the chatbot to deliver its features effectively and efficiently.

E. CHATBOT EVALUATION

To evaluate the performance of the proposed chatbot, we selected two complementary metrics: user satisfaction and response accuracy. These metrics were chosen based on a review of prior work, which shows that satisfaction/user assessment [32], [60], [61], [62], [63], [64] and accuracy evaluation [32], [65], [66], [67], [68] are prominent and recurrent in the development and evaluation of task-oriented and domain-specific chatbots. This dual approach enables both a subjective assessment from domain experts and a quantitative analysis based on ground-truth comparisons, providing a comprehensive understanding of the chatbot's capabilities and limitations.

1) SATISFACTION ASSESSMENT

The first evaluation approach focuses on measuring user satisfaction. For this purpose, a group of 18 users with expertise in Ecuadorian law participated in the evaluation. The methodology involved instructing each expert to pose 20 questions to the chatbot. Based on their expertise, the users rated their satisfaction with the chatbot's responses on a scale of 0 to 100, where 0 indicates responses that were entirely incorrect or irrelevant, while 100 represents responses that were precise, contextually appropriate, and accurate. To guide their evaluations, users were encouraged to consider aspects such as:

- **Correctness:** Whether the response accurately reflects the legal framework of Ecuador.
- **Relevance:** The degree to which the response is pertinent to the user's query.
- **Clarity:** How comprehensible and well-articulated the response is for the user.

To compute the overall satisfaction score, the average satisfaction rating across all questions and participants is

⁴<https://ollama.com/>

⁵<https://openwebui.com/>

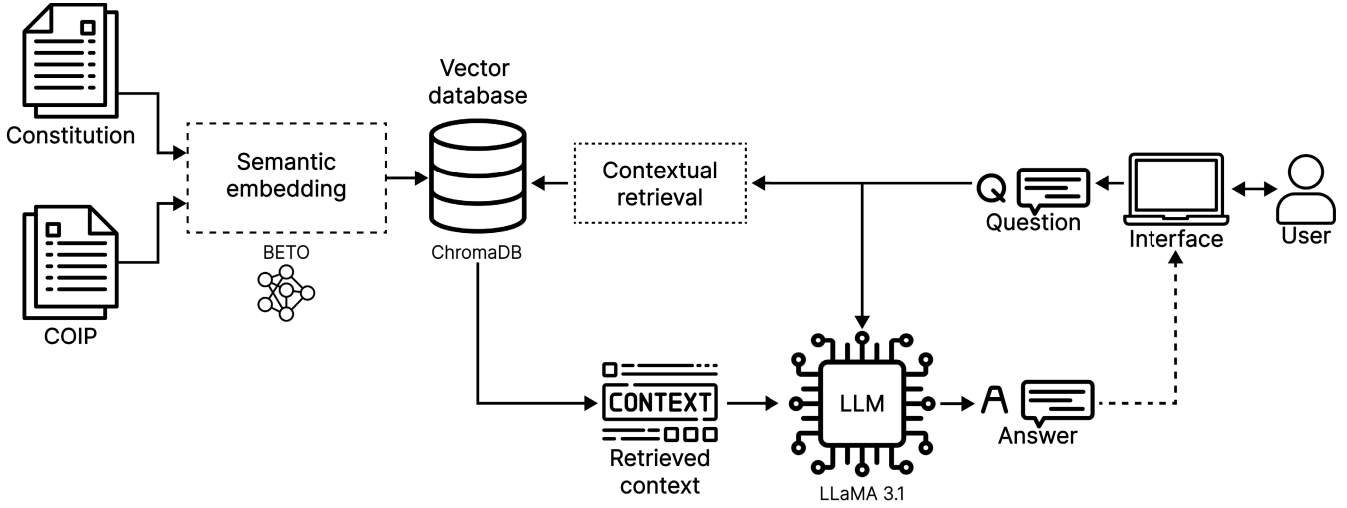


FIGURE 8. Flow diagram of the chatbot proposed in this work.

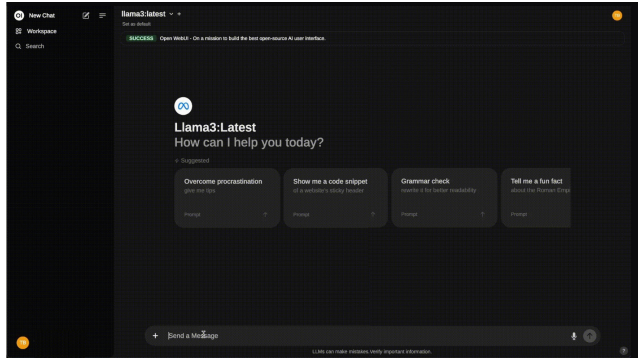


FIGURE 9. Example of the Open WebUI interface integrated with LLaMA.

calculated, as shown in Equation 1:

$$\text{Average satisfaction} = \frac{\sum_{i=1}^N s_i}{N}, \quad (1)$$

where s is the satisfaction score for question i provided by a user, and N is the total number of questions evaluated across all users. This score provides a quantitative measure of how well the chatbot meets the expectations of legal experts in providing accurate and user-friendly responses.

2) ACCURACY ASSESSMENT

While user-based satisfaction provides valuable insights into the chatbot's performance, it is inherently subjective and influenced by individual perspectives and expertise levels. This subjectivity can limit the ability to objectively measure the chatbot's accuracy across diverse queries. To address this limitation, a second evaluation approach is used, focusing on a quantitative analysis using a predefined dataset.

For this evaluation, a dataset is obtained with 25 questions and their corresponding ground-truths, derived directly from the legal documents used in this work. The questions are sourced from legal study materials, manuals, and question banks commonly used in Ecuadorian legal education. Each

question is passed to the chatbot, and the generated responses are compared against the ground-truth. The responses are classified into three categories:

- **Correct:** The response fully matches the ground-truth or paraphrases it without losing essential information.
- **Partially correct:** The response covers part of the intended answer but omits key information.
- **Incorrect:** The response fails to address the question accurately or provides erroneous information.

Each category is assigned a numerical weight, with correct responses receiving 1 point, partially correct responses 0.5 points, and incorrect responses 0 points. Using these weights, a weighted accuracy is calculated as shown in Equation 2:

$$\text{Weighted accuracy} = \frac{C + (0.5 \times P)}{N} \times 100, \quad (2)$$

where C is the number of correct responses, P is the number of partially correct responses, and N is the total number of questions evaluated.

This approach enables a deeper quantitative analysis of the chatbot's performance, accounting for nuanced responses that may arise in realistic scenarios. Furthermore, it facilitates the identification of error patterns and areas where the chatbot requires refinement, providing actionable insights for improving its functionality.

V. RESULTS AND ILLUSTRATIVE EXAMPLES

A. CHATBOT ASSESSMENT

To evaluate user satisfaction, a testing session was conducted at the University of Otavalo, Ecuador's simulated courtroom with students in their final semester of the Law degree, enrolled in the Criminal Litigation course. A total of 18 participants posed 20 questions each to the chatbot and evaluated its responses. The questions covered general topics related to the COIP and the Ecuadorian Constitution. Each response was rated on a satisfaction scale from 0 to 100. For each

participant, an individual satisfaction score was computed by averaging the ratings of their 20 responses. Finally, the overall satisfaction level was obtained by averaging the individual scores across all participants.

The results of this evaluation are presented in Table 1. From the table, it can be observed that the chatbot achieved a high overall satisfaction score of 88.72, indicating a generally positive reception among participants. The majority of users rated their experience above 85, with only a few scores at the lower end of the range (80). Notably, one participant rated the chatbot's performance with the maximum score of 100, suggesting that, in some cases, the system provided responses perceived as fully accurate and useful. These results suggest that the chatbot is effective in delivering legally relevant and comprehensible responses. In addition, participants noted that the chatbot is a potentially valuable tool as a resource to support their studies. However, the variation in scores also implies that there is room for improvement, particularly in ensuring consistency across different queries and user expectations.

TABLE 1. Chatbot satisfaction evaluation results.

Participant	Satisfaction score
1	80
2	85
3	90
4	85
5	80
6	90
7	100
8	80
9	95
10	88
11	80
12	90
13	99
14	90
15	90
16	85
17	95
18	95
Average	88.72

To provide a clearer understanding of the satisfaction distribution, Figure 10 presents a histogram of the collected ratings. The histogram illustrates that the chatbot received predominantly high ratings, with most scores clustering between 80 and 100. The distribution appears slightly right-skewed, indicating that while the majority of users rated their experience favorably, a few ratings were slightly lower. Notably, the highest frequency is observed around the 90-95 range, suggesting that most users found the chatbot to be highly satisfactory. The density curve further reinforces this trend, showing a peak in the upper range of the scale. These results indicate that the chatbot generally meets user expectations in providing relevant and accurate legal responses, though some variation in individual satisfaction levels is present.

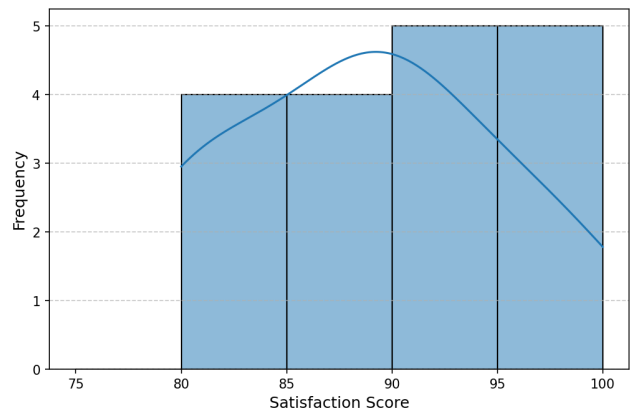


FIGURE 10. Distribution of satisfaction scores.

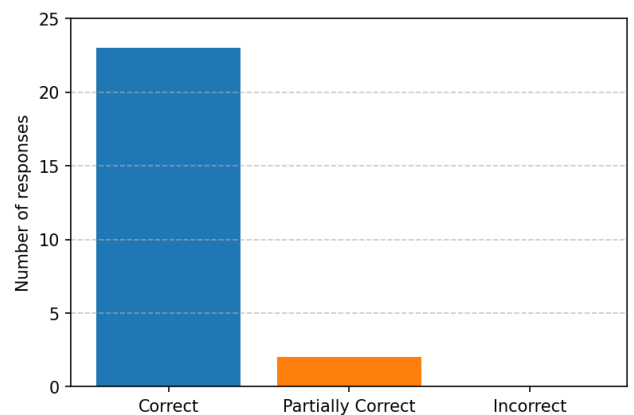


FIGURE 11. Distribution of accuracy assessment chatbot response categories.

Iniciar sesión en Open WebUI

Email

Contraseña

Iniciar sesión

FIGURE 12. Chatbot login screen.

For the accuracy assessment, each response generated by the chatbot was compared against the predefined ground-truth derived from legal documents. The evaluation was conducted independently by each of the authors of this research, who classified responses as correct, partially correct, or incorrect based on the criteria outlined in Section IV-E2. If all authors independently assigned the same classification to a response, it was directly accepted. In cases where discrepancies arose, a secondary review was conducted collaboratively to reach a



FIGURE 13. Chatbot home screen.

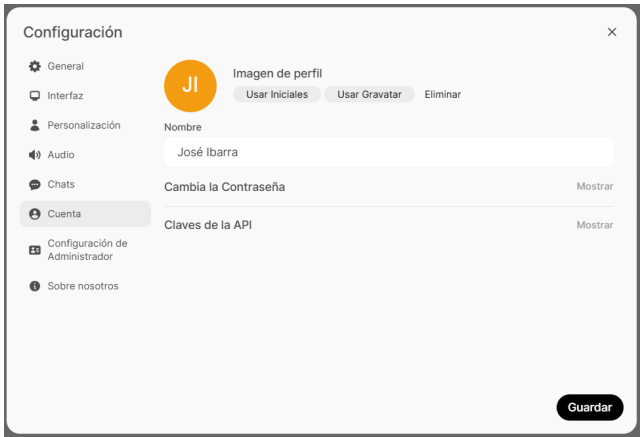


FIGURE 14. User profile management interface.

consensus on the final classification. This approach ensured a rigorous and unbiased evaluation of the chatbot’s accuracy. The results of the accuracy assessment are presented in Figure 11. A total of 23 responses were classified as correct, 2 as partially correct, and none were categorized as incorrect. This indicates that the chatbot provides responses that are largely aligned with the expected ground-truth, with only minimal deviations in certain cases. Numerically, the derived weighted accuracy from these results is 96%, demonstrating that the chatbot is highly reliable in generating legally accurate responses. The presence of a small percentage (4%) of partially correct responses suggests that, while the chatbot performs well, there are occasional instances where relevant legal details are omitted or not fully articulated. These cases highlight areas where further refinements in retrieval or response generation could enhance precision. The dataset containing the evaluation questions, ground-truth answers, and chatbot-generated responses is provided as

supplementary material, allowing for independent verification of the chatbot’s performance as reported in this study.

B. ILLUSTRATIVE EXAMPLES AND CHATBOT FEATURES

Since the chatbot utilizes Open WebUI as its front-end interface, it inherits many of the features provided by this tool. Additionally, as the chatbot is specifically designed for Ecuadorian legal queries, its default language is set to Spanish. Upon accessing the system, users are presented with the option to log into the chat or register as a new user, as shown in Fig. 12.

Once logged in, the interface closely resembles other LLM-based chatbots, such as ChatGPT. As shown in Fig. 13, the main screen consists of a simple chat interface with a central input bar where users can enter their queries. The interface also includes options for voice input and additional settings, ensuring an intuitive and accessible user experience.

The chatbot also provides a configuration panel where users can personalize their experience. As shown in Fig. 14, the settings menu includes options for managing the account, such as changing the display name, updating credentials, and configuring API keys. Additionally, users can adjust interface preferences, enable or disable audio features, and access administrator settings if they have the necessary permissions. These customization options enhance usability and allow for a more tailored interaction with the system.

Additionally, on the administration side, there are various functionalities available for configuring the chatbot. Among these, one of the most important is the ability to manage the knowledge base, as shown in Fig. 15. Administrators can upload new files, synchronize directories, or manually add textual content. This functionality allows for expanding the chatbot’s knowledge base beyond the initially integrated legal documents, enabling the incorporation of additional laws, regulations, or case-specific materials as needed.

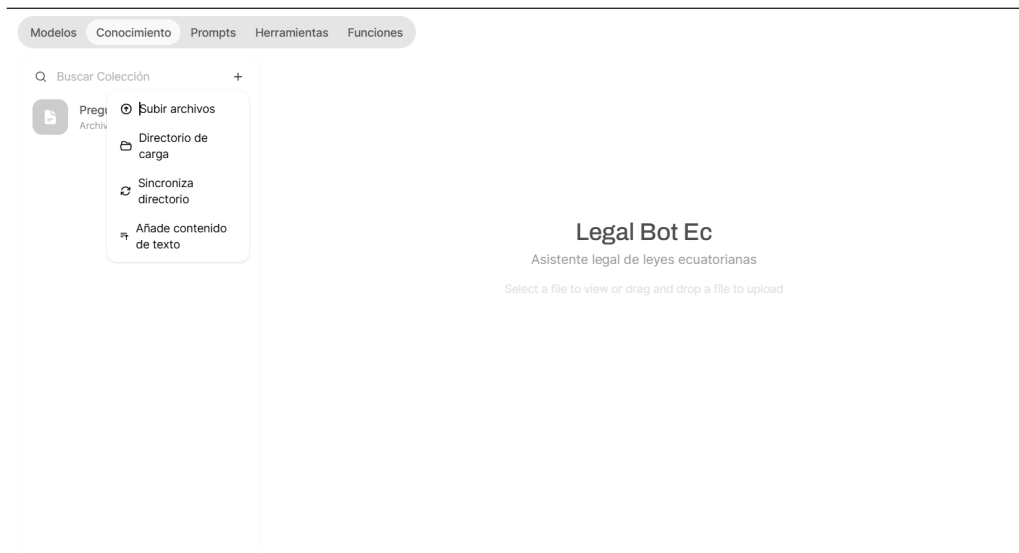


FIGURE 15. Knowledge base management interface.

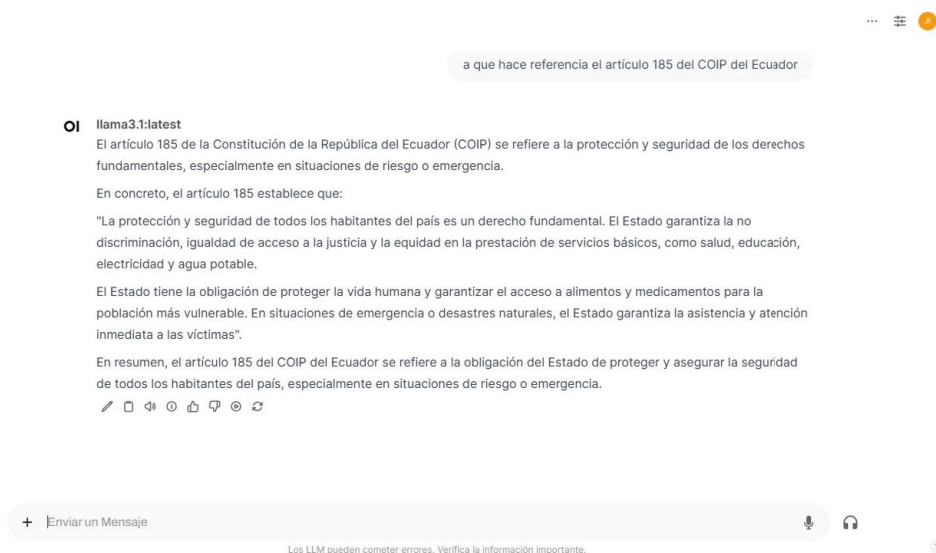


FIGURE 16. Example of how the chatbot works.

Now, regarding its functionality, Fig. 16 presents an example of a user query and the chatbot's response. In this case, the chatbot was asked about the reference of Article 185 of the COIP in Ecuador. The response correctly summarizes the article's content, highlighting its focus on the protection and security of fundamental rights, particularly in emergency situations. The chatbot structures the answer clearly, outlining key points related to state obligations, non-discrimination, and access to essential services. Additionally, it reinforces reliability by explicitly referencing the legal source used in its response.

Finally, Fig. 17 presents an additional test evaluating the chatbot's multilingual capabilities. In this case, a question was posed in English regarding whether a Justice of the Peace can make decisions that override Indigenous Justice.

The chatbot successfully interprets the query and provides a comprehensive response. It explains the recognition of Indigenous Justice within Ecuador's legal framework, specifically referencing Article 171 of the 2008 Constitution. Additionally, the chatbot contextualizes the role and jurisdiction of Justices of the Peace, citing Article 190, and outlines the limitations and scope of both legal systems. This response demonstrates the chatbot's ability to handle multilingual inputs while maintaining accurate legal references.

In summary, the results from both the satisfaction and accuracy assessments, along with the functional evaluations, demonstrate that the proposed chatbot effectively provides legally grounded responses based on Ecuadorian law. The system exhibits high accuracy in its answers, as well as strong user satisfaction, indicating its potential as a reliable tool

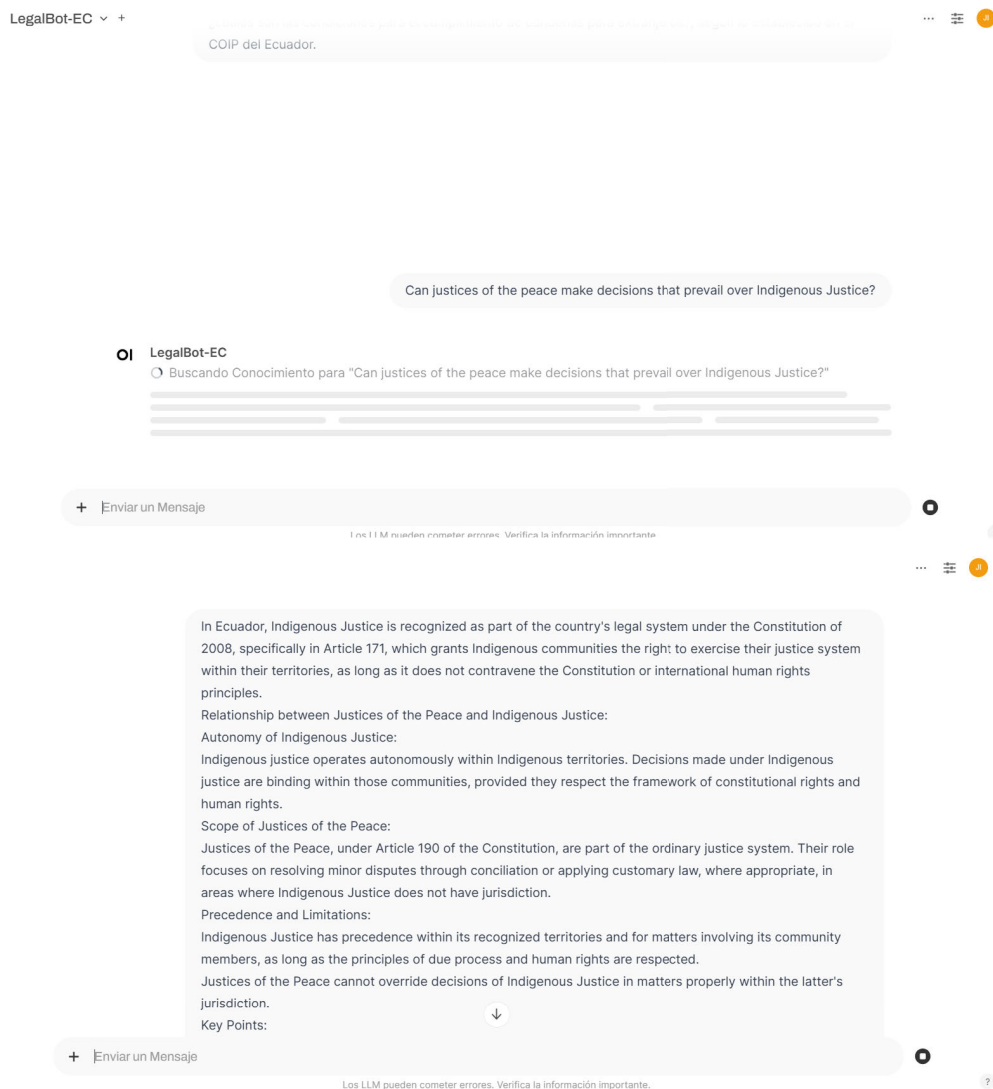


FIGURE 17. Example of how the chatbot works with a question in English. Above, the question asked, below, the chatbot's response.

for legal information retrieval. Additionally, the functionality tests confirm that the chatbot operates smoothly within its designed framework, supporting interactions in both Spanish and English while correctly leveraging its knowledge base.

VI. ETHICAL CONSIDERATIONS AND BIAS MITIGATION

The development and deployment of LegalBot-EC raise several ethical considerations, particularly due to its role in processing and delivering legal information to users. While the system is designed as a support tool for accessing legal content, care must be taken to ensure its responsible use and interpretation.

A. USE LIMITATIONS AND USER GUIDANCE

LegalBot-EC is not intended to replace qualified legal professionals. Although it has been evaluated by domain experts and grounded in official legal sources, it should be understood as a complementary tool aimed at facilitating

access to relevant legal information. Users are advised not to treat its outputs as definitive legal advice, particularly in contexts involving legal procedures, court decisions, or legal obligations.

B. BIAS IN LANGUAGE MODELS AND LEGAL SOURCES

The system relies on large pre-trained language models, which may carry biases present in the data they were trained on. While LegalBot-EC mitigates this to some extent through retrieval-based grounding from official legal texts, the underlying language models may still reflect social, cultural, or institutional biases inherent in general pretraining corpora. Moreover, the legal documents used represent a specific interpretation of the Ecuadorian legal framework. While authoritative, these sources do not encompass all legal domains or regional legal practices. This limitation must be acknowledged, especially when users expect exhaustive or highly specific responses.

C. BIAS MITIGATION AND VALIDATION EFFORTS

To reduce the risk of hallucinations and support faithful generation, LegalBot-EC employs a RAG approach. This ensures that answers are grounded in content extracted from the legal documents used, rather than relying purely on the generative model's prior knowledge. All responses are thus contextually constrained to Ecuadorian law. Additionally, the system was evaluated by a group of final-year law students with training in criminal litigation. Their feedback, collected through a structured evaluation protocol, serves not only to validate the quality of the chatbot's answers but also to identify areas where legal nuance or clarity may be lacking. This human-in-the-loop process helps reduce the impact of potential errors or oversights.

D. TRANSPARENCY AND REPRODUCIBILITY

The development of LegalBot-EC relies exclusively on publicly available tools and models. BETO was obtained from the Hugging Face model hub, and LLaMA 3.1 was integrated via local deployment through Ollama. The retrieval system is implemented using ChromaDB. All processing steps, from text extraction to embedding generation and answer synthesis, are fully documented and reproducible.

VII. LIMITATIONS

The proposed chatbot, while effective in its current implementation, presents some limitations. Its knowledge base is restricted to two foundational legal documents, the COIP and the 2008 Constitution, which, although comprehensive, do not encompass the entirety of Ecuador's legal framework. This limitation affects its ability to address queries related to other laws, regulations, or localized legal contexts. Additionally, the system's reliance on preprocessed text means that errors during cleaning or segmentation could negatively impact response accuracy. While LLaMA 3.1's enriched responses enhance its functionality, the model may struggle with complex queries that require integrating information from multiple dispersed sections or rely heavily on its pre-trained knowledge when retrieved fragments are insufficient. Furthermore, as additional legal documents are integrated, ensuring scalability and maintaining efficiency in retrieval and response generation will be critical challenges.

Another important consideration is the methodology used for accuracy assessment. The evaluation was conducted by the authors themselves following a cross-validation approach, where responses were independently categorized before reaching a consensus. Although this method was intended to ensure fairness and consistency, and although the dataset containing the chatbot responses has been included as supplementary material for the sake of transparency and verification, the absence of external evaluators may introduce a potential source of bias. Further assessments involving independent experts and expanded datasets would help strengthen the reliability and generalizability of the chatbot's performance.

VIII. CONCLUSION AND FUTURE WORK

In this work, we propose the development of a legal chatbot designed to provide responses grounded in Ecuadorian law by leveraging LLMs. The chatbot integrates ChromaDB for contextual retrieval and LLaMA 3.1 as its core generative model, ensuring responses are both legally relevant and contextually enriched. Its knowledge base consists of two foundational legal documents: the COIP and the 2008 Constitution of Ecuador, which serve as the primary sources for generating responses. To facilitate user interaction, the system is deployed using Open WebUI, offering an accessible and intuitive interface.

The evaluation of the proposed chatbot was conducted through two key assessments: user satisfaction and accuracy measurement. The satisfaction assessment, performed with final-year law students at the University of Otavalo, Ecuador, yielded an average satisfaction score of 88.72 out of 100, indicating a high level of approval regarding the chatbot's correctness, relevance, and clarity. Meanwhile, the accuracy assessment, based on a benchmark of 25 legal questions with predefined ground-truth answers, demonstrated strong performance. The chatbot correctly answered 23 questions, provided partially correct responses for 2 questions, and had no incorrect responses, resulting in a weighted accuracy of 96%. Additionally, illustrative examples of chatbot interactions confirmed its ability to provide coherent and well-grounded responses, even when queried in English, demonstrating basic multilingual support.

Future research can build upon this work expanding the chatbot's knowledge base by incorporating additional legal documents beyond the COIP and the Ecuadorian Constitution, allowing it to handle a broader range of legal inquiries. Further evaluations with legal professionals could refine the system's accuracy and usability in practical applications. Additionally, testing the chatbot with more complex legal scenarios would provide insight into its strengths and limitations when handling intricate legal reasoning. Beyond this specific implementation, similar chatbots could be developed for other legal frameworks in other countries. Furthermore, the approach could extend beyond the legal domain to fields such as medicine or finance.

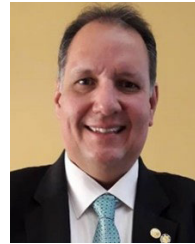
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